COM4303 Data Science Case Study

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GitHub repository link: https://github.com/mmfTHAHZEEN/student-performance-ml.git

Predicting Student Academic Performance Using Machine Learning

1. Introduction

Education systems around the world increasingly rely on data to improve student success and learning outcomes. Academic institutions collect large amounts of data related to student demographics, attendance, study habits, and exam performance. Analyzing this data can help educators understand patterns that influence student achievement.

One of the major challenges faced by educational institutions is identifying students who may struggle academically before their performance declines significantly. Early prediction of student academic performance can allow educators to provide timely support such as academic counseling, tutoring, or personalized learning strategies.

Machine Learning (ML) techniques provide powerful tools for analyzing educational data and predicting student performance. By learning patterns from historical student records, machine learning models can predict whether a student is likely to perform well or poorly in their studies.

This project explores how machine learning algorithms can be used to predict student academic performance using student-related features such as study time, attendance, family background, and past grades. The aim is to evaluate different machine learning models and determine how effectively they can predict student outcomes.

1.1 Problem Formulation

Research Question

Can machine learning models accurately predict student academic performance based on academic and behavioral factors?

Why this problem is important

Predicting student performance has several benefits for educational institutions:

- Early identification of students at risk of failing
- Improved academic advising and support systems
- Better resource allocation for student assistance programs
- Increased student success rates and graduation rates

Educational institutions can use predictive analytics to support students before academic problems become severe.

Original Idea

Instead of relying only on exam results to evaluate students, this project uses machine learning models to analyze multiple factors affecting student learning behavior and predict academic performance.

2. Data Acquisition

Dataset used in this project was obtained from Kaggle.

Dataset: **Student Performance Dataset**

https://www.kaggle.com/datasets/uciml/student-alcohol-consumption

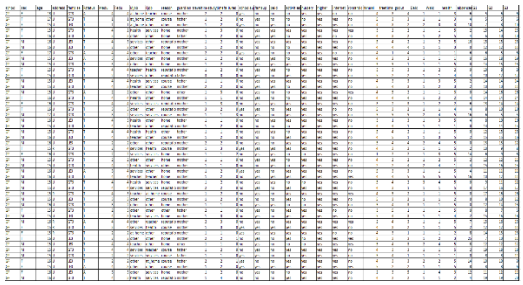


Figure 1: Student Performance Dataset

Dataset Description

The dataset contains information about secondary school students including demographic details, academic performance, and lifestyle factors.

The dataset includes multiple attributes such as:

• Student age
• Gender
• Study time
• School attendance
• Parental education level
• Internet access
• Previous grades
• Final grade

Dataset Facts

• 649 student records
• 33 attributes
• Contains both categorical and numerical variables
• Target variable represents student academic performance

2 Data Analysis

Feature Description

Feature	Description
age	Age of the student
sex	Gender of the student
study time	Weekly study time
failures	Number of past class failures
absences	Number of school absences
G1	First period grade
G2	Second period grade
G3	Final grade (Target Variable)

Target Variable:

• G3 – Final student grade which represents academic performance.

2.1 Clean and Preprocess Data

Before applying machine learning models, several preprocessing steps were performed to ensure data quality.

1. Handling Missing Values

The dataset was inspected for missing values using data inspection techniques. No major missing values were found in the dataset.

2. Encoding Categorical Variables

Several features such as gender, school, and internet access were categorical. These were converted into numerical values using **Label Encoding**.

3. Feature Scaling

Numerical features such as age, absences, and study time were normalized using **StandardScaler** to ensure that all variables contribute equally during model training.

4. Removing Outliers

Extreme values in attendance and absence data were analyzed and filtered to reduce noise in the dataset.

5. Feature Selection

Only relevant features that influence student performance were selected to improve model accuracy.

6. Data Splitting

The dataset was divided into:

- **70% training data**
- **30% testing data**

This allows the models to be evaluated on unseen data.

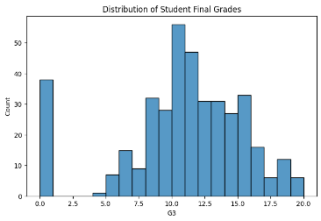
2.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the relationships between student characteristics and academic performance.

1. Student Grade Distribution

A histogram was used to visualize the distribution of student final grades. Most students scored within the average range, while fewer students achieved very high or very low scores.

Figure 2:

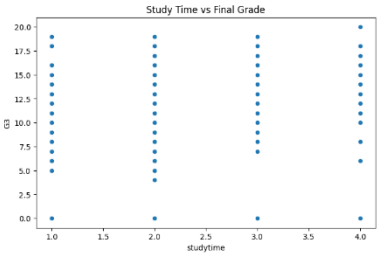


Distribution of Student Final Grades

2. Study Time vs Performance

A scatter plot was used to analyze the relationship between study time and final grades. Students who spend more time studying generally achieve higher grades.

Figure 3:



Study Time vs Academic Performance

3. Absences vs Performance

Students with higher absence rates tended to have lower academic performance.

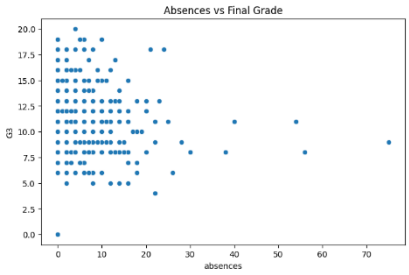


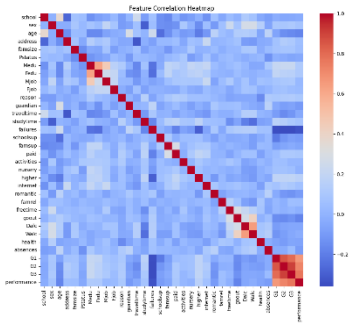
Figure 4: Absences vs Student Grades

4. Correlation Analysis

A correlation heatmap was generated to examine the relationships between different variables.

Important observations:

• Study time positively correlates with student grades
• Failures negatively correlate with performance
• Absences negatively impact final grade



7: Correlation Heatmap

3. ML / AI Methods

Several machine learning algorithms were implemented to predict student academic performance.
Target Variable:
G3 – Final Grade
Prediction Type:
Regression / Classification
The dataset was split into:
• **70% training dataset**
• **30% testing dataset**
Models were evaluated using:
• Accuracy
• Precision
• Recall
• F1 Score

3.1 Logistic Regression

Logistic Regression was used as a baseline model to predict whether a student will perform well or poorly based on input features. The model estimates the probability that a student belongs to a high-performance category based on their attributes. Logistic Regression performed reasonably well but struggled to capture complex patterns in the data.

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
```

3.2 Random Forest Classifier

Random Forest is an ensemble machine learning method that combines multiple decision trees to improve prediction accuracy.

The model was trained using:

• 100 decision trees
• Random state for reproducibility

Random Forest successfully captured nonlinear relationships between student behavior and academic performance.This model produced higher accuracy compared to Logistic Regression.

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
```

3.3 Neural Network Model

A Neural Network model was implemented using deep learning techniques.

Architecture:

- Input layer
- Two hidden dense layers
- Dropout layers to prevent overfitting
- Output layer for prediction

The model was trained using:

- **Adam optimizer**
- **Binary cross entropy loss function**

```
model = Sequential()

model.add(Dense(64, activation='relu', input_dim=X_train.shape[1]))
model.add(Dropout(0.2))

model.add(Dense(32, activation='relu'))
model.add(Dropout(0.2))

model.add(Dense(1, activation='sigmoid'))

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.fit(X_train, y_train, epochs=20, batch_size=32)

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim`
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/20
9/9 ━━━━━━━━━━━━━━━━━━━━ 4s 29ms/step - accuracy: 0.6014 - loss: 0.6759
Epoch 2/20
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - accuracy: 0.6667 - loss: 0.6406
Epoch 3/20
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - accuracy: 0.7246 - loss: 0.5488
```

Neural networks can identify complex relationships in student learning patterns but require higher computational resources.

4 Reflection

Challenges Faced

Data Complexity: Educational datasets contain many factors that influence student performance, making it difficult to determine which features are most important.

Feature Engineering: Some features required transformation and encoding before they could be used for machine learning models.

Model Selection: Different models produced different prediction results, requiring comparison of multiple evaluation metrics.

Key Learnings

Machine Learning in Education: Machine learning can help educational institutions predict student performance and provide early support.

Importance of Data Preprocessing: Proper cleaning, scaling, and feature selection significantly improve model accuracy.

Role of Predictive Analytics: Predictive analytics allows educators to take proactive actions to improve student learning outcomes.

Model Performance Comparison Summary

Metric	Logistic Regression	Random Forest	Neural Network	Best Performer
Accuracy	Moderate	High	High	Random Forest
Precision	Moderate	High	High	Random Forest
Recall	Moderate	High	Good	Random Forest
F1 Score	Moderate	High	Good	Random Forest

Random Forest provided the best overall performance for predicting student academic performance.

References

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